

Knowledge Retrieval Enhancement for Power System Based on Adaptive Dynamic Text Segmentation and Hierarchical Multi-Query

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Abstract—To address the issue of long document correlation information retrieval and understanding in complex power system data, this paper proposes a power system knowledge retrieval enhancement method based on dynamic text segmentation and hierarchical multiple queries. An adaptive dynamic text segmentation method is proposed, which dynamically adjusts the window size and stride according to the text content to adapt to the dynamic segmentation of text under different information densities. A hierarchical multiple query mechanism is proposed, which decomposes the original problem into multiple sub-problem queries, and improves the depth and breadth of retrieval by reasoning and integrating the answers to each sub-problem, ensuring the comprehensiveness and accuracy of retrieval. Experiments are conducted on six datasets, and the results showed that the F1 scores under two vector databases (Chroma and Milvus) are 52.52 and 55.85 respectively, indicating that the proposed method can effectively enhance the power system knowledge retrieval and understanding capabilities, providing technical references for further optimization and prediction of power systems.

Index Terms—Power system, Large language models, Retrieval Augmented generation, Knowledge enhancement

I. INTRODUCTION

The global energy transition towards low-carbon economies has led to an exponential increase in complex, heterogeneous

data generated by power systems, encompassing equipment status, grid operation logs, and maintenance records [1]. These multi-domain datasets provide critical support for optimizing grid dispatch and predictive analytics. However, the high dimensionality and heterogeneity of power system data pose significant challenges for efficient integration and analysis, necessitating advanced computational frameworks to enable intelligent management and decision-making in power systems.

Recent advances in large language models (LLMs) have demonstrated remarkable capabilities in semantic understanding and large-scale data processing, particularly in long-document processing and knowledge-based question answering [2], [3]. With context windows exceeding 200,000 tokens, LLMs offer improved accuracy and coherence in handling extended texts. Nevertheless, their application to power system data is hindered by limitations in semantic precision (e.g., disambiguating domain-specific terminology), contextual coherence (e.g., maintaining long-range dependencies in operation logs), and domain adaptability (e.g., incorporating power system-specific knowledge). These challenges underscore the need for tailored solutions to enhance LLMs' performance in power system applications.

To address these limitations, retrieval-augmented generation (RAG) has emerged as a promising approach by integrating external knowledge retrieval to enhance the reliability and relevance of LLM outputs [3], [4]. In power system data analysis,

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RAG facilitates semantic integration and contextual analysis, mitigating biases in single-query responses and supporting complex scenarios such as fault diagnosis and dispatch optimization. However, the effectiveness of RAG heavily relies on the relevance and accuracy of retrieved information. Inaccurate or irrelevant retrievals can significantly degrade the quality of generated outputs, compromising the reliability of power system analytics.

To overcome these challenges, this study proposes two innovative techniques: (1) **Adaptive Dynamic Text Segmentation (ADTextSeg)**, which dynamically adjusts sliding window size and step length to accommodate the complex structure and semantic density of power system texts, preserving contextual integrity while minimizing information loss; and (2) **Hierarchical Multi-Query (HMQ)**, which generates multi-level, semantically related query variants to enhance the retrieval of comprehensive and accurate information from external sources. By combining ADTextSeg with HMQ, this study aims to improve the retrieval precision and generation quality of RAG systems, thereby enabling robust data analysis and decision support for power system applications.

II. RELATED WORK

A. Multi-Domain Data in Power Systems

The increasing complexity of power system data, driven by the global energy transition, necessitates advanced data-driven approaches for grid optimization and predictive analytics. Cheng et al. [5] proposed a data-driven method for smart grid optimization, leveraging multi-domain data to enhance dispatch and forecasting accuracy. Sahu et al. [6] developed a mechanism integrating real-time multi-domain data to improve situational awareness in power systems. Similarly, Zhao et al. [7] introduced the Multi-popo framework, which utilizes multi-timescale data to improve load forecasting precision. These studies highlight the critical role of multi-domain data integration, motivating the need for robust computational frameworks to handle their heterogeneity.

B. Large Language Models (LLMs)

Large language models (LLMs) have gained attention for their potential in power system applications due to their advanced semantic understanding. Pribylskii [8] reviewed LLMs' applications in power system operation and control, emphasizing their roles in dispatch optimization and user interaction. Pathak et al. [9] surveyed AI techniques, including LLMs, for power flow control and voltage stability, underscoring the importance of data availability and interpretability. Jia et al. [10] proposed a feedback-driven multi-agent framework that integrates retrieval-augmented generation (RAG) and error feedback mechanisms, significantly enhancing LLMs' performance in complex simulation tasks. Ruan et al. [11] demonstrated the reliability of LLM agents in power system control using Pandapower and LLaMA 3. These advancements pave the way for applying LLMs to multi-domain power system data, though challenges in domain-specific adaptability remain.

C. Retrieval-Augmented Generation (RAG)

Retrieval-augmented generation (RAG) enhances LLMs by integrating external knowledge retrieval, improving the accuracy of code generation and task execution in power system simulations. Lewis et al. [3] introduced the original RAG framework, segmenting Wikipedia documents into chunks and embedding them in a vector space for efficient semantic retrieval. However, this approach struggles to capture power system-specific terminology and dependencies. Gao et al. [12] categorized RAG into naive, advanced, and modular paradigms, highlighting the need for continuous optimization in retrieval components. Optimization strategies include retrieval source, indexing, and query enhancements [13]. In indexing, document preprocessing quality directly impacts retrieval accuracy, requiring structured segmentation and semantic embedding. Fixed-size chunking (e.g., 100/256/512 tokens) reduces noise but risks semantic truncation [14]. To address this, dynamic segmentation based on semantic similarity was proposed, which can effectively avoid semantically related content being segmented into different parts [15]. This study builds on these findings by proposing a sliding window strategy with adaptive overlap to enhance text coherence.

Query optimization improves RAG's ability to handle complex intents in power systems. Ma et al. [16] proposed query rewriting to mitigate semantic mismatches between queries and documents. Dhuliawala et al. [17] introduced a "chain-of-verification" method to validate expanded queries, enhancing reliability. Despite these advances, incomplete queries persist, prompting this study to leverage LLMs for multi-perspective query reconstruction, reducing information loss and improving retrieval precision.

III. METHOD

A. Problem Definition

In the field of power systems, large language models (LLMs) integrated with retrieval-augmented generation (RAG) techniques offer new possibilities for processing complex technical documents and real-time data. However, power system text data, such as equipment operation logs, grid dispatch reports, and technical specifications, exhibit multi-domain characteristics, dense domain-specific terminology, and long-context dependencies, posing unique challenges for corpus construction and index optimization. To effectively support RAG systems in power system applications, a high-quality corpus training set T must be constructed to ensure retrieval accuracy and domain relevance of generated content.

1) *Objectives of Corpus Construction:* The construction of the corpus training set T aims to transform unstructured power system text data (e.g., operation logs, dispatch instructions, and maintenance records) into structured information units, enabling efficient semantic retrieval and generation. This study takes power system text data D as input, with the goal of dividing D into a series of semantically coherent chunks $\{W_1, W_2, \dots, W_n\}$ through optimized segmentation and indexing methods, stored in a vector database, to achieve the following objectives:

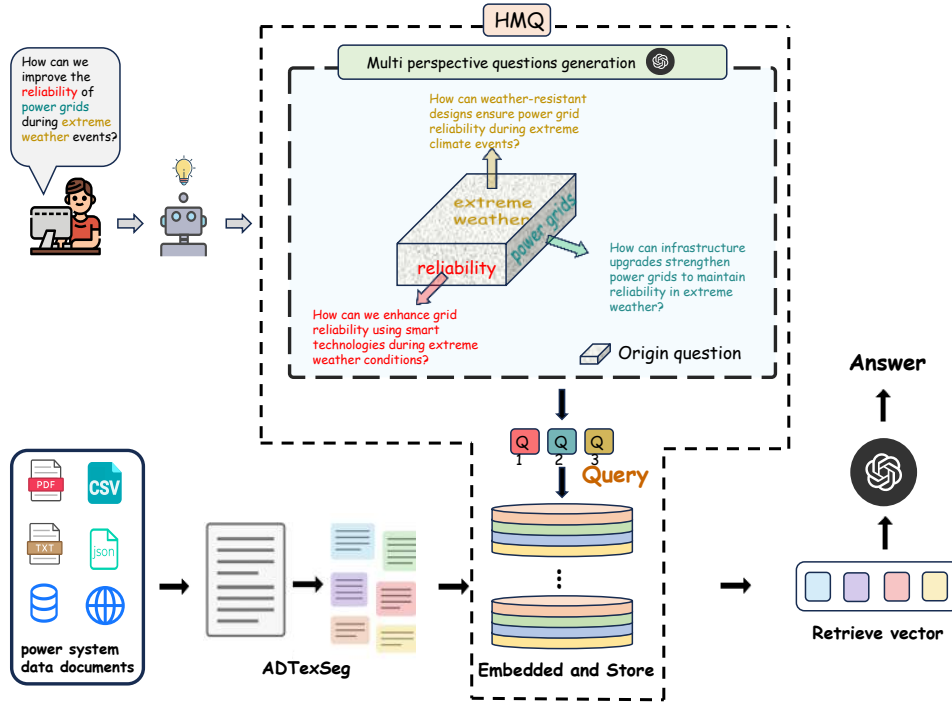


Fig. 1. Pipeline for Processing Power System Corpora with ADTextSeg and HMQ

- **Semantic Coherence:** Ensure each chunk W_i preserves critical contextual information in power system texts (e.g., equipment status descriptions, fault analysis, or dispatch logic), avoiding the truncation of semantic units during segmentation.
- **Information Completeness:** Easily capture domain-specific information in D (e.g. technical terminology, time series data, and grid topology), ensuring that critical content is not omitted in T .
- **Domain Adaptability:** Enhance the adaptability of corpus T to multi-domain power system data (e.g., text, tables, and time-series), meeting the needs of diverse application scenarios such as fault diagnosis and dispatch optimization.

2) *Mathematical Formulation:* Given a power system text dataset D with length L (measured in tokens), the goal of corpus construction is to segment D into n chunks $\{W_1, W_2, \dots, W_n\}$ using a function f , where w denotes the window size and s denotes the sliding step. This is formally expressed as:

$$[W_1, W_2, \dots, W_n] = f(D, w, s) \quad (1)$$

The number of chunks n can be approximated by:

$$n = \left\lceil \frac{L - w}{s} \right\rceil + 1 \quad (2)$$

Each chunk W_i is defined as:

$$W_i = D[(i - 1) \times s : \min((i - 1) \times s + w, L)] \quad (3)$$

Here, $\lceil x \rceil$ denotes the ceiling function and $\min()$ ensures that the last window does not exceed the length of D . The

core problem lies in selecting an appropriate segmentation strategy, optimizing w and s to balance semantic coherence and information coverage while accommodating the complex structure of power system texts (e.g., multilevel technical descriptions and dynamically updated data streams).

3) *Challenges and Innovations:* Existing corpus construction methods, such as fixed-size segmentation, face the following challenges when processing power system text data:

- Fixed-window segmentation can truncate technical terms or critical contexts, reducing semantic integrity.
- General-purpose embedding models struggle to capture domain-specific knowledge in power systems, which affects retrieval accuracy.
- The heterogeneity of multi-domain data increases the complexity of corpus construction.

This study proposes an improved corpus construction method, combining a sliding window strategy with domain-adaptive embedding generation, to optimize the construction of T , thus improving retrieval efficiency and generation quality for RAG systems in power system applications.

B. Overall Pipeline

ADTextSeg addresses common limitations of fixed-window and static retrieval methods in power system data processing, mitigating context loss and retrieval inaccuracies. The system architecture comprises key components that enhance the generation of accurate responses from large language models (LLMs). As shown in Fig. 1, the workflow begins with collecting multi-domain power system corpora, including PDFs, CSV files, and plain text from equipment logs and

grid operation records. These serve as private data sources for information retrieval.

Documents undergo preprocessing to identify semantic units such as sentences and paragraphs, ensuring segmentation preserves semantic boundaries. ADTextSeg then applies a sliding window technique to divide text into overlapping blocks, which are indexed and embedded into a vector database. This indexing approach enables fine-grained, contextually accurate retrieval, enhancing LLMs' ability to generate reliable and coherent responses for power system applications. The implementation process includes:

- **Window Segmentation:** This step dynamically analyzes the semantic structure of power system corpora, adjusting window size and step length based on content markers like punctuation and paragraph breaks. By flexibly setting window sizes, the system retains complete semantic units within each text segment, improving contextual relevance and information integrity for grid data analysis.
- **Embedding and Indexing:** Each segmented window is processed by a pre-trained language model to generate vector representations, which are stored in a vector database. This ensures efficient storage and precise retrieval of power system-specific embeddings, supporting complex query handling.
- **Pre-Retrieval:** Leveraging MultiQueryRetriever technology, this phase embeds queries into a distance-based vector space for retrieval. However, semantic mismatches between queries and embeddings may yield inconsistent results, often requiring manual adjustments or prompt engineering, which can be inefficient. To address this, the system integrates an advanced feature where LLMs automatically generate multiple variants of user queries, enhancing retrieval by capturing diverse perspectives.

Building on the limitations of pre-retrieval, the Hierarchical Multi-Query (HMQ) mechanism further refines this process by introducing a structured approach to query optimization. This method addresses the challenge of incomplete semantic coverage in power system corpora, such as equipment status logs and grid operation data, by generating multiple query variants. By leveraging these variants, HMQ enhances the retrieval of comprehensive and contextually relevant information, laying a robust foundation for the subsequent stages of analysis and response generation:

- **Retrieval:** Upon receiving multiple query variants, the system accesses the vector database using advanced retrieval algorithms to fetch relevant paragraphs. By fusing results from all queries, it forms a comprehensive document set, improving coverage and capturing subtle differences in power system contexts.
- **Response Generation:** Retrieved text segments are input as context for LLMs, which synthesize the information to produce coherent, fact-based responses. Leveraging extensive training data and provided context, LLMs deliver accurate solutions tailored to power system challenges.

C. Adaptive Dynamic Text Segmentation

Power system documentation data often exhibits characteristics such as lengthy text content, dispersed data, and highly variable information density. To effectively process textual information while maximizing key information capture and ensuring semantic coherence, this study proposes an **Adaptive Dynamic Text Segmentation Strategy (ADTextSeg)**. By dynamically adjusting the segmentation window size and step length based on textual content, the method addresses the contextual understanding constraints caused by fixed token limits in models when processing long texts.

Let D denote a document with length L , and s represent the text segmentation step size. Under the proposed ADTextSeg, both the window size and step length are dynamically adjusted according to equation 4, where W_i denotes the i -th segmentation window extracted from document D . The starting position of the window is $(i - 1) \times s$, and the segmentation span is determined jointly by the window size and the starting position's special tokens, while ensuring the window does not exceed the total text length L . The i -th window can be expressed as:

$$W_i = D[(i - 1) \times s + \text{next}(D, (i - 1) \times s, T) : \min((i - 1) \times s + w + \text{next}(D, (i - 1) \times s, T), L)] \quad (4)$$

Algorithm 1 presents the pseudocode for ADTextSeg. First, the text D is partitioned using special token T to identify important paragraphs. Then, through dynamically adjusting window size and stride, it iteratively creates windows W_i based on content and special token positions, achieving adaptive text segmentation under varying information densities.

Algorithm 1 Adaptive Dynamic Text Segmentation Strategy

Input: Text D ; Specific symbol T ; Window size w ; Step size s

Output: Text Segmentation Window W

```

1:  $W = []$ ; start = 0
2: splits = split_text( $D, T$ )
3: while start < len(splits) do
4:   index = start; total_length = 0; temp_chunk = ''
5:   while index < len(splits) do
6:     total_length = total_length + len(splits[index])
7:     if total_length  $\geq w$  then
8:        $W.append(temp\_chunk)$ 
9:       break;
10:    end if
11:    index = index + 1;
12:  end while
13:  start = next(splits, start, s, T)
14: end while
15: return  $W$ 
```

D. Hierarchical Multi-Query

The complexity of power systems typically involves numerous variables and dynamic processes, making it challenging for single queries to fully capture all relevant factors. To

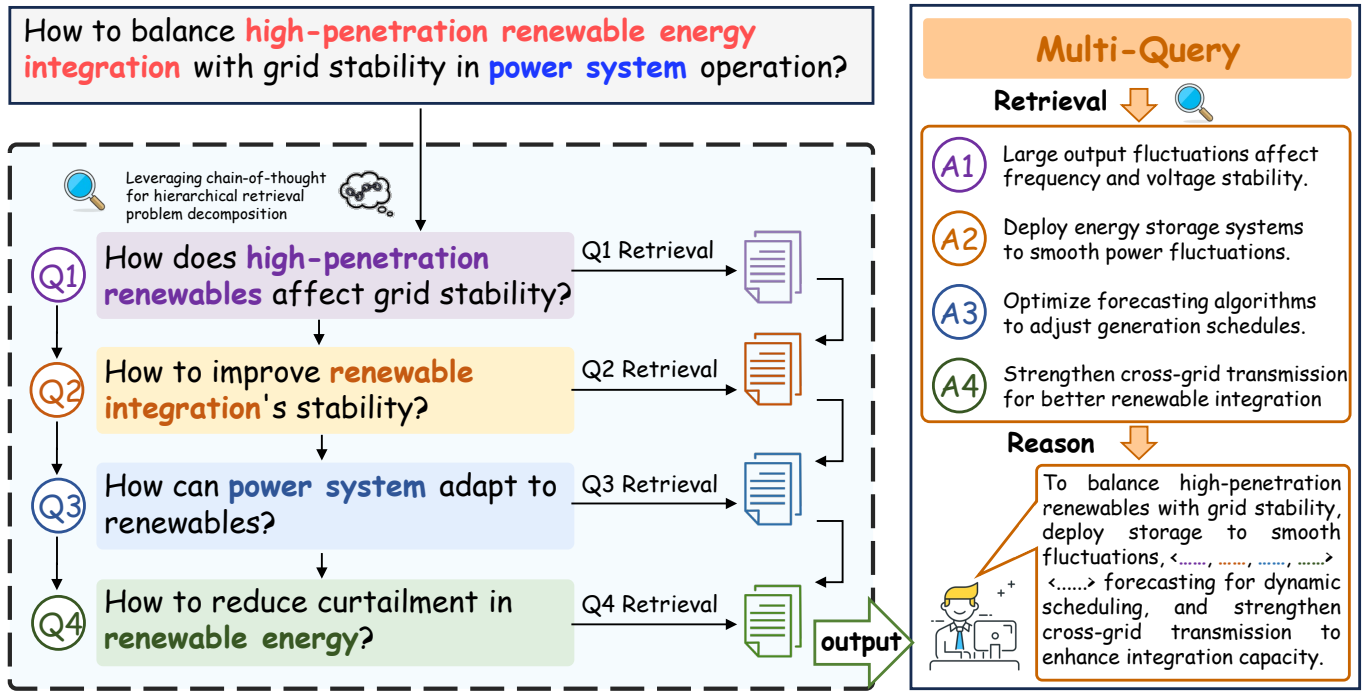


Fig. 2. Hierarchical Multi-Query in Power System Q&A

address this, we propose an innovative **Hierarchical Multi-Query (HMQ)** method (shown in Fig. 2). This approach significantly enhances the precision and efficiency of retrieval by decomposing the original user query into multiple sub-queries. Specifically, upon receiving a user's query, the system performs retrieval for each sub-query by accessing the most relevant document segments in the vector database, ultimately integrating all query results through comprehensive reasoning to form the final answer. This method not only deepens and broadens the retrieval scope but also ensures the comprehensiveness and accuracy of the answers.

Taking the query "How to balance high-penetration renewable energy with grid stability" as an example, we first identify the key factors affecting grid stability and design corresponding sub-questions for each factor. Through hierarchical retrieval, we can obtain detailed answers for each sub-question and use them as input for the next level of retrieval. Specifically, we break down the question into four sub-questions: "How do high-penetration renewables affect grid stability?", "How to improve the stability of renewable integration?", "How can the power system adapt to renewables?", and "How to reduce curtailment in renewable energy?". For each sub-question, the system uses efficient retrieval algorithms to find the most relevant documents and integrates the retrieval results of all sub-questions to form a comprehensive answer. By providing multiple perspectives on the same issue, this technology not only enhances comprehensive coverage of the topic but also improves the ability to capture subtle differences. This method is not only applicable to the field of power systems but can also be extended to other areas that require

complex problem decomposition and resolution.

IV. EXPERIMENT

A. Implement Details

Dataset. To investigate the retrieval performance of the proposed method based on ADTextSeg and HMQ for enhancing knowledge retrieval in power systems across various domains, this study conducts experimental validation on the publicly available datasets NarrativeQA [18], Qasper [19], and HotpotQA [20]. Furthermore, experiments are specifically conducted on three typical document types in power system: maintenance records (**PowerDoc-MR**), operation reports (**PowerDoc-OR**), and regulatory standards (**PowerDoc-RS**). Table II provides detailed information on the six datasets, where the number of queries refers to the count of queries in the test set, and the number of documents indicates the quantity of documents within the corpus.

Experiment settings. This study employs LLaMA-3 [21] as the large language model for retrieval augmented generation of power system knowledge, using the *LLaMA-3-chatembeddings* as the embedding model. The RAG prompt used is as follows: "Answer the question based solely on the following context. Context: {context}, Question: {question}"

Evaluation metrics. As for the evaluation of retrieval performance, we adopt the F1 score as the metric for the results of RAG knowledge retrieval, which provides a balanced measure of precision and recall. In this project, *Answerpred* denotes the set of output results from our method, while *Answercorrect* denotes the set of reference answers. Precision and recall are calculated according to equations

TABLE I
COMPARISON OF F1 SCORES ACROSS DIFFERENT DATASETS

Method	Public Dataset			Power Dataset			Avg
	NarrativeQA	Qasper	HotpotQA	PowerDoc-MR	PowerDoc-OR	PowerDoc-RS	
Chroma (baseline)	60.11	57.03	53.21	33.73	42.76	32.55	45.57
Chroma (+ADTextSeg)	62.76	59.74	52.44	36.36	42.89	34.84	48.17
Chroma (+HMQ)	63.81	53.42	60.47	35.72	43.26	36.75	48.91
Chroma (+ADTextSeg+HMQ)	69.74	62.68	59.21	38.19	45.69	39.63	52.52
Milvus (baseline)	64.14	63.35	46.35	42.83	50.26	41.37	51.38
Milvus (+ADTextSeg)	64.27	64.47	49.24	42.12	52.13	46.62	53.14
Milvus (+HMQ)	67.56	63.50	45.73	43.21	49.81	47.35	52.86
Milvus (+ADTextSeg+HMQ)	71.38	65.78	51.82	44.19	52.47	49.48	55.85

TABLE II
DATASET DETAILS

Dataset	Domain	Token-len(Avg)	Language	Num-files
NarrativeQA [18]	Literature	18409	English	200
Qasper [19]	Science	3619	English	200
HotpotQA [20]	Wikipedia	9151	English	200
PowerDoc-MR	Power	4559	Chinese	324
PowerDoc-OR	Power	6701	Chinese	241
PowerDoc-RS	Power	27534	Chinese	23

$$Precision = \frac{|Answerpred \cap Answercorrect|}{|Answerpred|} \quad (5)$$

$$Recall = \frac{|Answerpred \cap Answercorrect|}{|Answercorrect|} \quad (6)$$

Therefore, the F1 score can be further expressed as:

$$F1 = 2 \times \frac{Precision \times Recall}{Precision + Recall} \quad (7)$$

B. Experimental Results

The vector database supports the storage of vectorized representations of content in the knowledge base. Chroma [22] employs the maximum marginal relevance algorithm for document selection, which optimizes both the relevance of queries and the diversity of documents. Milvus [23] is widely used for efficient similarity search in large-scale vector datasets. The Recursive Character TextSplitter (RCS) splits text by prioritizing the natural boundaries of paragraphs and sentences. In the baseline model, with LLaMA-3 serving as the language question-answering model, Chroma and Milvus are selected as the vector databases for retrieval, and RCS is used as the text splitting method. Experiments are conducted on both public datasets and power system datasets to verify the effectiveness of the proposed ADTextSeg and HMQ in retrieval enhancement, and the results are shown in table I.

With Chroma and Milvus as vector databases, our proposed retrieval enhancement methods achieved average F1 scores of **52.52** and **55.85** across six datasets, respectively, outperforming other methods. Although our method did not achieve the best performance on the HotpotQA dataset when using Chroma as the vector database, it still significantly outperformed the baseline methods. Given the complexity and high professionalism of the information in power system data documents, the F1 scores on the power datasets were generally

low. Even so, our proposed method still achieved a substantial improvement over the baseline models, further demonstrating its strong potential for understanding and retrieving complex power system documents.

V. CONCLUSION

In this study, we proposed a novel approach to enhance knowledge retrieval in power systems by introducing two innovative techniques: Adaptive Dynamic Text Segmentation (ADTextSeg) and Hierarchical Multi-Query (HMQ). The primary objective of this research was to address the challenges associated with retrieving and understanding correlated information from long, heterogeneous documents within multi-domain power system data. ADTextSeg tackles the issue of varying information densities by dynamically adjusting window size and stride based on text content, ensuring semantic coherence and minimizing information loss during segmentation. Meanwhile, HMQ enhances retrieval comprehensiveness and accuracy by decomposing complex queries into multiple sub-queries, broadening the scope and depth of retrieved information through integrated reasoning.

Experimental evaluations were conducted across six datasets, comprising both public benchmarks (NarrativeQA, Qasper, HotpotQA) and power system-specific corpora (PowerDoc-MR, PowerDoc-OR, PowerDoc-RS). The results demonstrated the superior performance of our proposed method, achieving average F1 scores of 52.52 and 55.85 with Chroma and Milvus vector databases, respectively. These scores reflect significant improvements over baseline methods, underscoring the effectiveness of ADTextSeg and HMQ in enhancing retrieval precision and generation quality. Notably, the method exhibited robust adaptability to the complex, domain-specific nature of power system data, providing a reliable foundation for applications such as fault diagnosis, grid optimization, and predictive analytics.

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